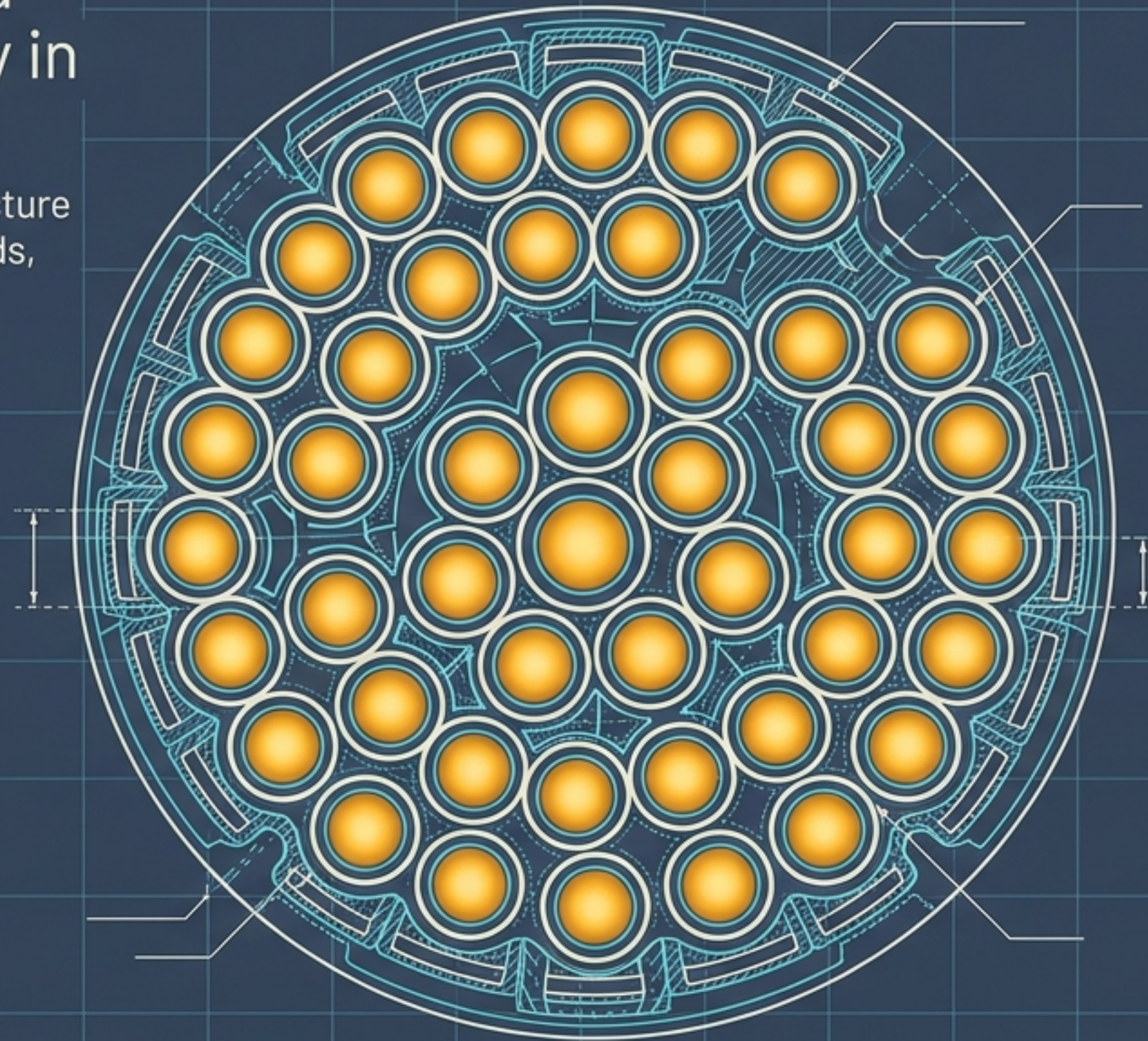


Phase Fraction and Filament Geometry in Structured Foods

A Configurable Process Architecture for Emulsions, Core-Shell Strands, and Consolidated Textures



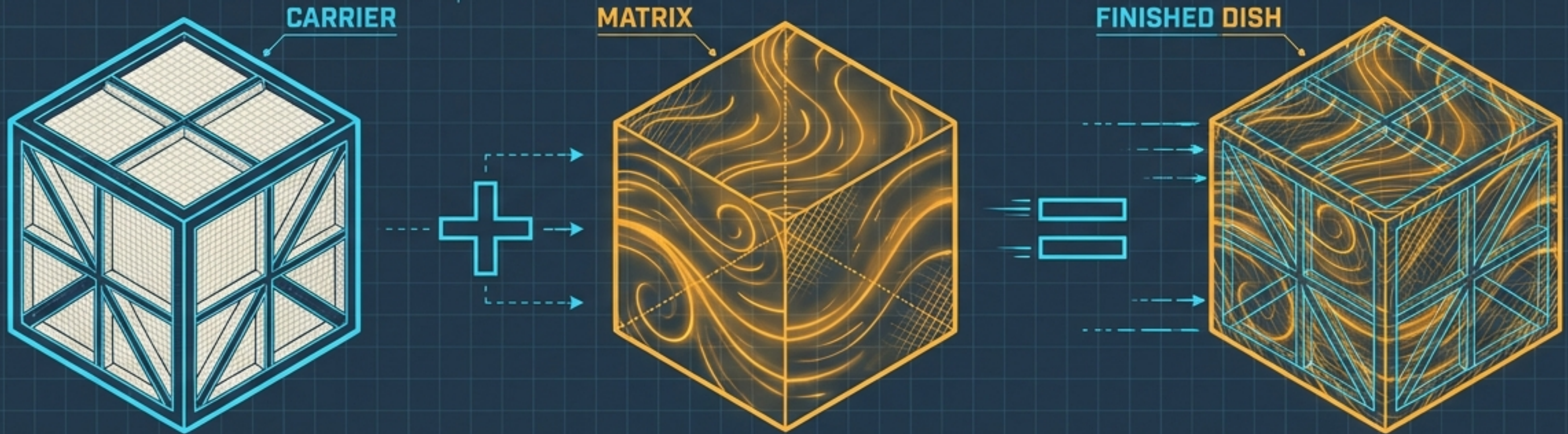
0.00mm

5.00mm

DECONSTRUCTING DISH IDENTITY

Hypothesis 1 (Late Binding): Beyond a threshold of flavor complexity, the perceived identity of a savory dish is determined by its final aroma, sauce, and fat profile. Meatiness is a reaction class (Maillard, sulfur, lipid degradation) applied during finishing, not an inherent requirement of the structure.

[CARRIER (STRUCTURE)] + [MATRIX (FLAVOR/FAT)] + [INCLUSIONS] = DISH IDENTITY

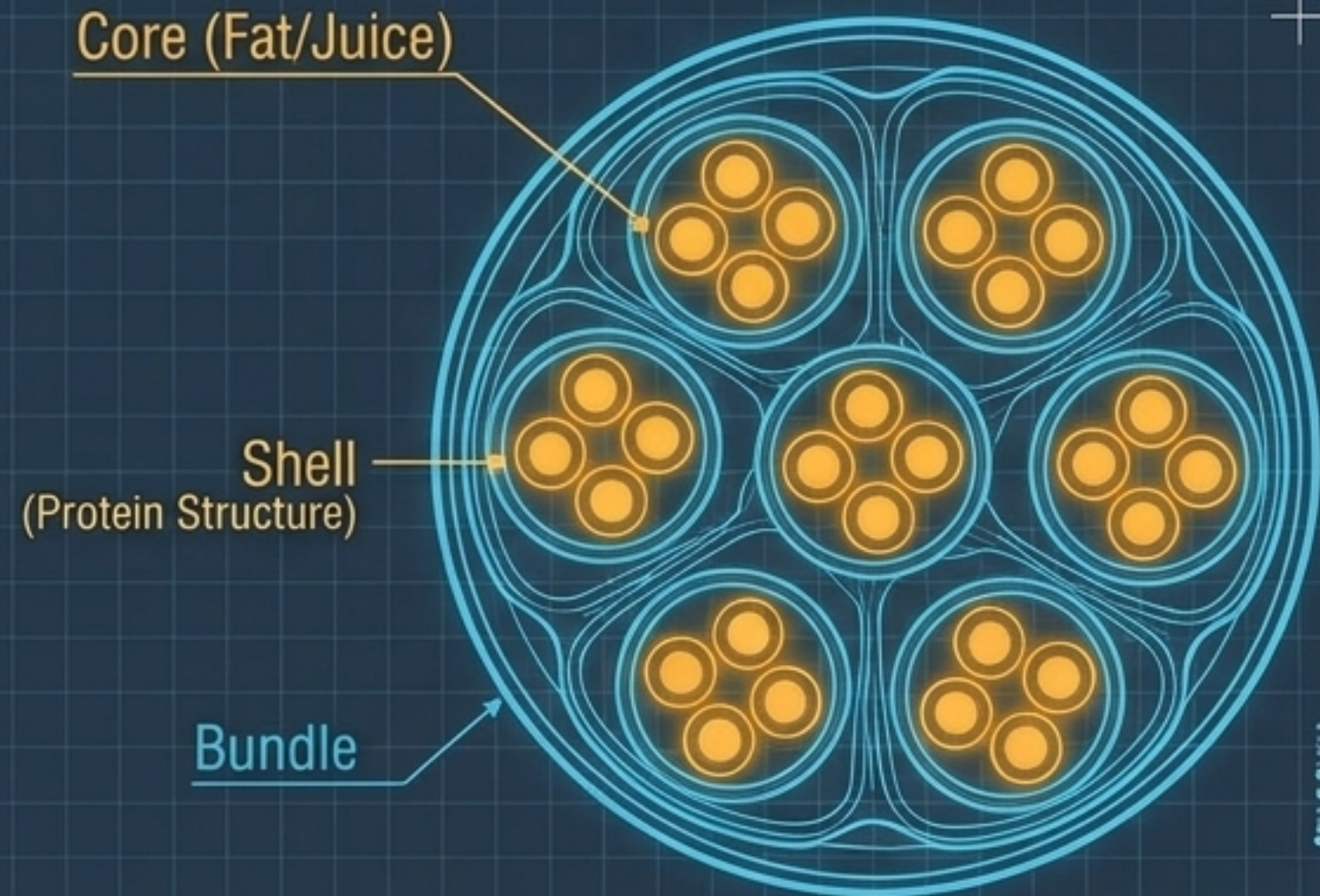
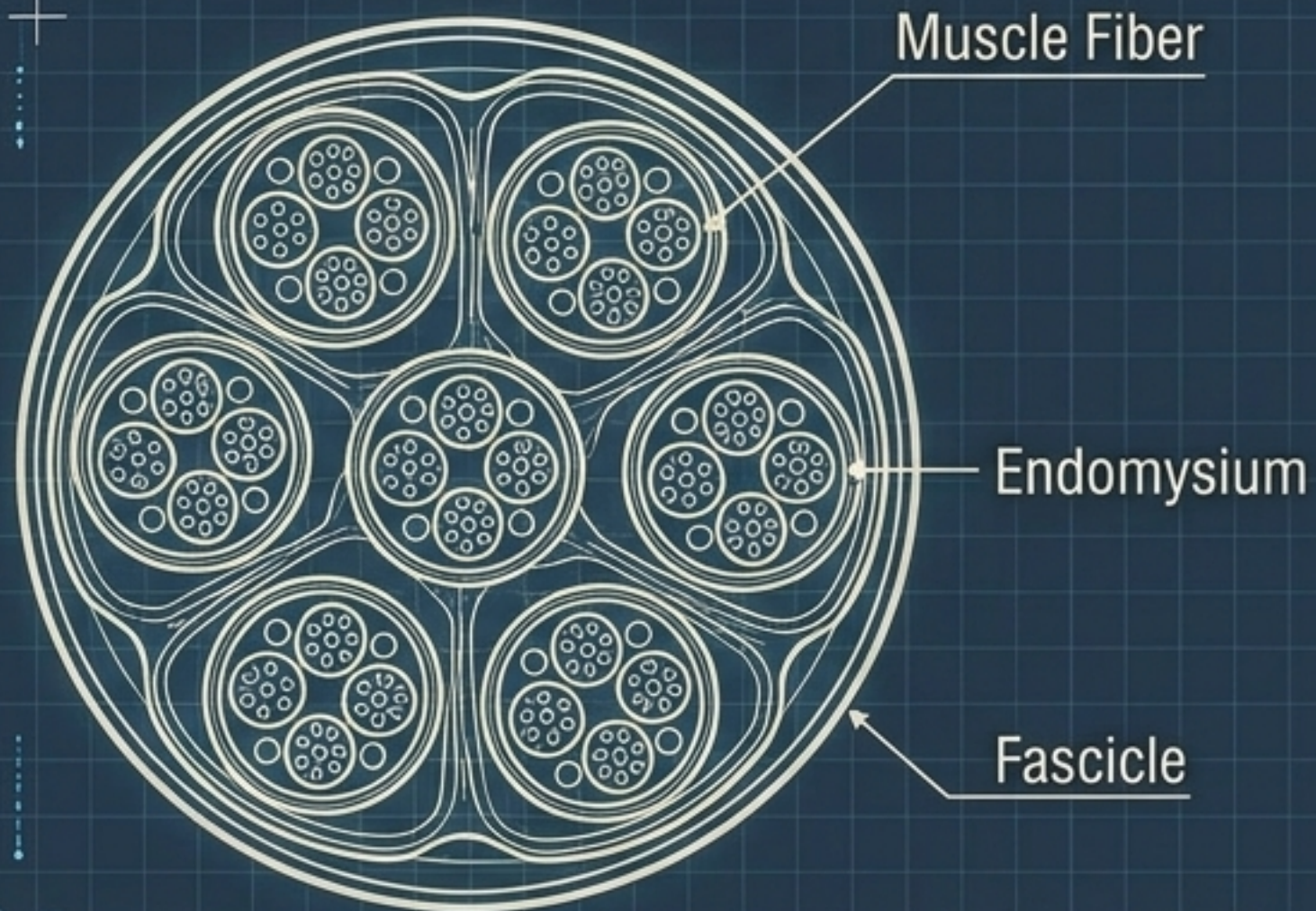


The Structuring Technology Landscape

Existing methods either control micro-alignment at the expense of phase placement, or control placement at the expense of bulk speed. Assembly solves this.

Method	What it controls	What it leaves uncontrolled	Output type
High-Moisture Extrusion	Alignment from flow	Phase placement	Continuous mass
Shear Cells	Alignment under defined shear	Multiphase arrangement	Batch block
Printing	Exact placement	Fine fiber speed	Slow layered build
Cultivated Tissue	Biological structure	Cost and scale	Biological tissue
Core-Shell Strands (Proposed)	Phase fraction (ϕ), diameter (d), orientation (θ), bonding (f)		Configurable blocks

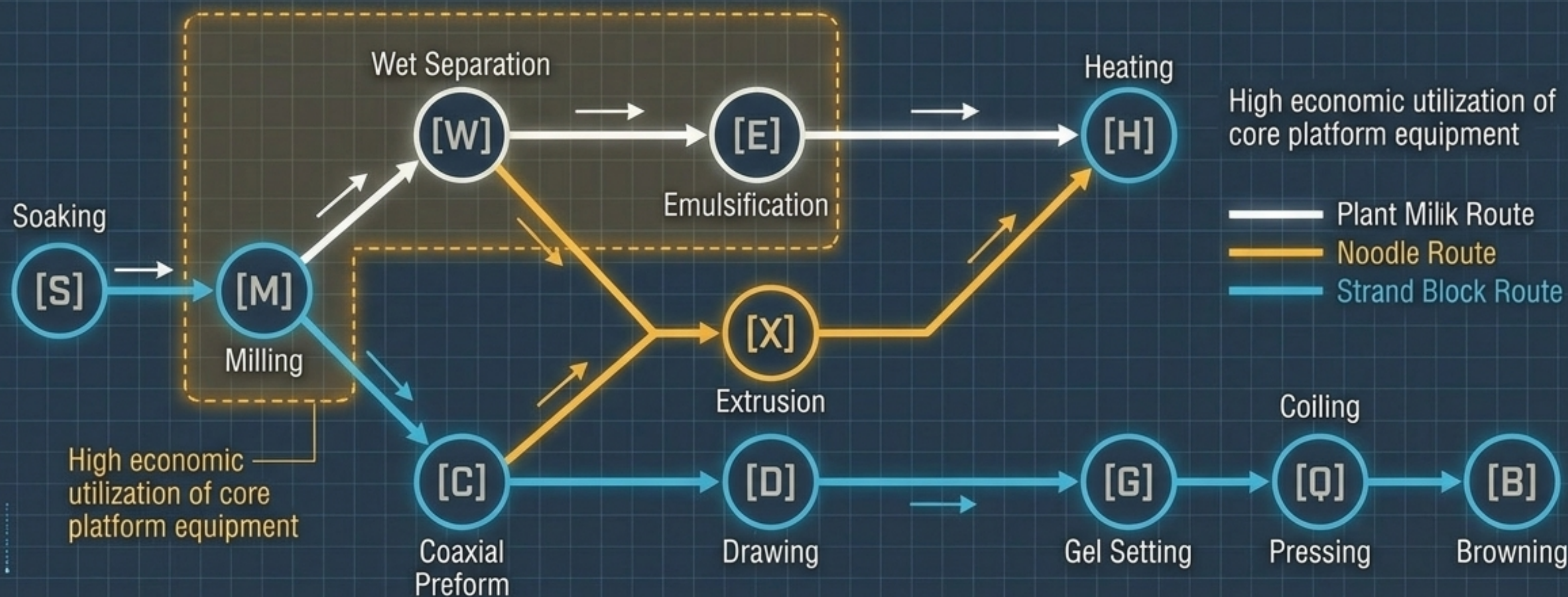
Biological Blueprints vs. Engineered Architecture



Muscle is a reference structure, not a target. It provides a precedent for sheathed strands in a fused network associated with grain, shred, and juiciness.

The Operator Graph: Products as Configurable Routes

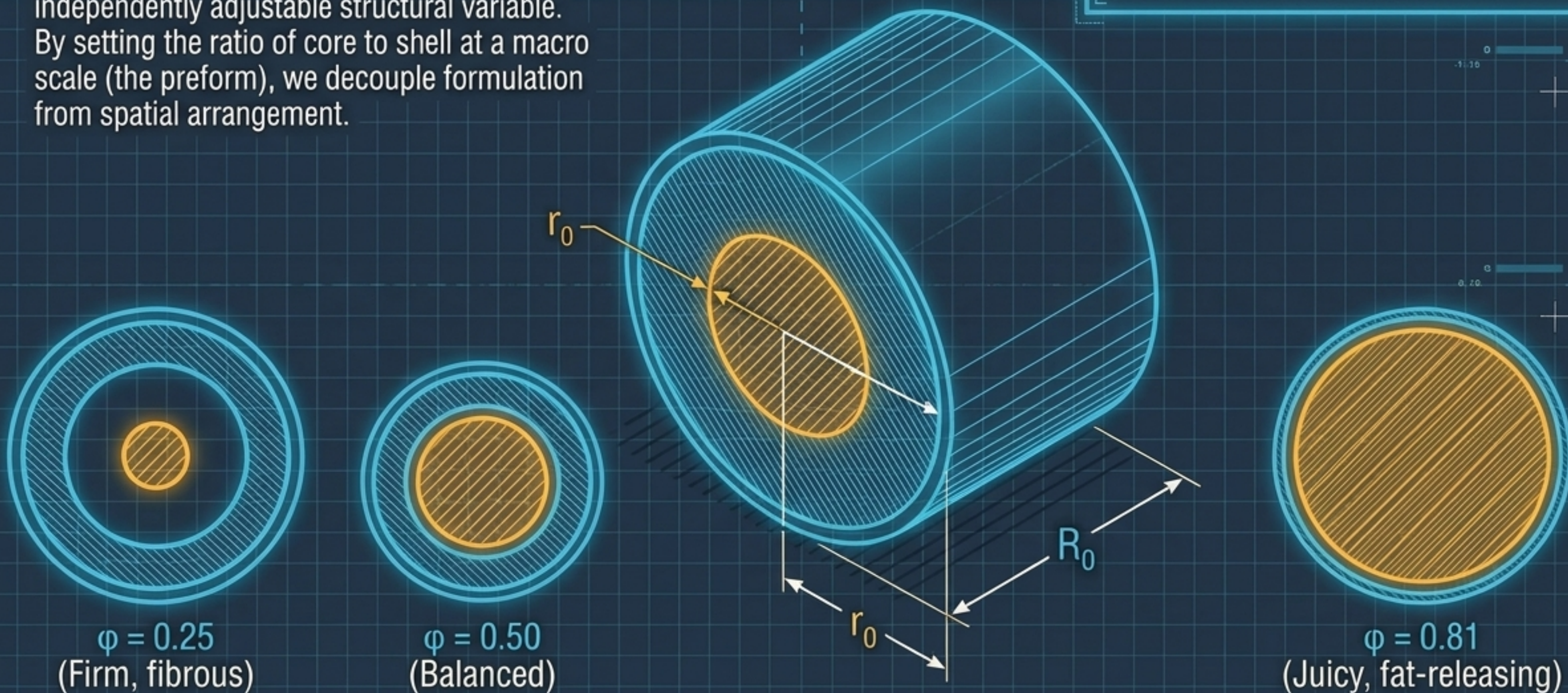
The platform separates material intake from product identity. A product is simply a programmed route through shared fractionation, transformation, and forming operators.



Anatomy of the Preform: Phase Fraction (ϕ)

- Hypothesis 2:** The proportion of the inner phase is an independently adjustable structural variable. By setting the ratio of core to shell at a macro scale (the preform), we decouple formulation from spatial arrangement.

$$\text{Core Area Fraction } (\phi) = \left(\frac{r_0}{R_0} \right)^2$$



Setting the Shell: The Engineering Trade-off

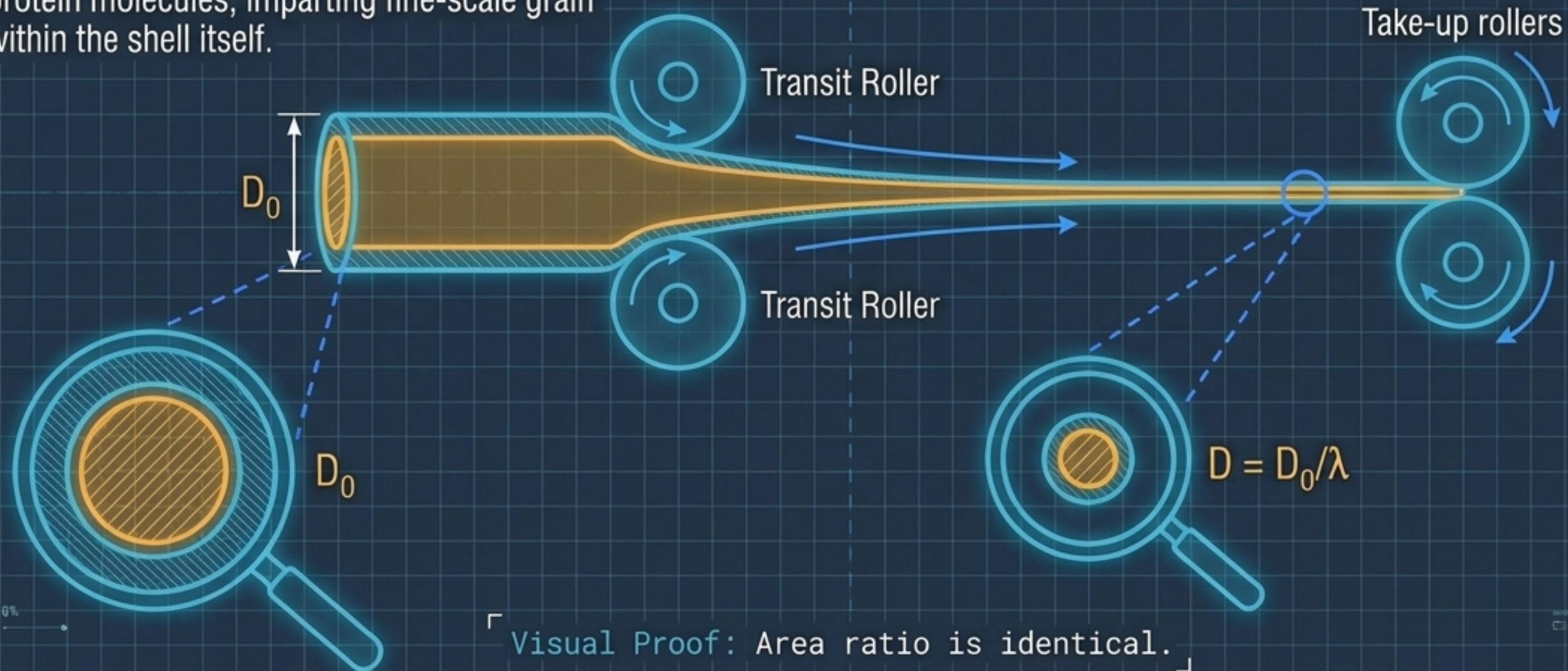
The shell must contain the core during drawing, survive bending during coiling, and still chemically fuse to its neighbors during final pressing.

Setting Route	Containment	Flexibility	Bondability
Thermal	Good	Uncertain (may be stiff)	Reduced by full set
Ionic (Alginate)	Good	Good	Poor without binder
Frying	Strongest	Good hot, poor cold	Likely poor
Drying	Good	Moderate	Recoverable by remoistening
Enzymatic	Slow to develop	Good early	Good if pressed in time



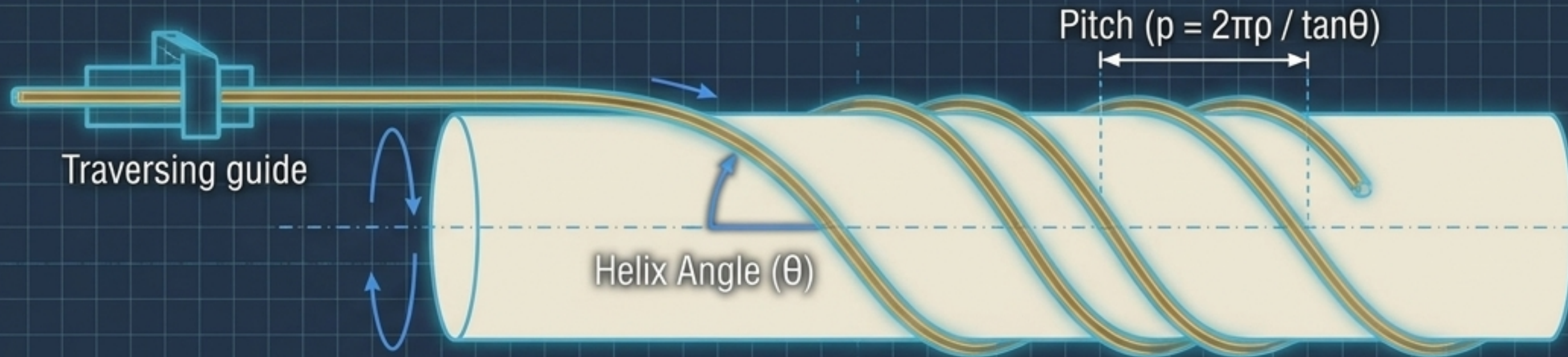
Draw Invariance: Scaling from Macro to Micro

- Hypothesis 3 (Draw Invariance): A core-shell preform can be drawn to a much smaller diameter while mathematically preserving its core fraction (ϕ). Drawing also aligns long protein molecules, imparting fine-scale grain within the shell itself.

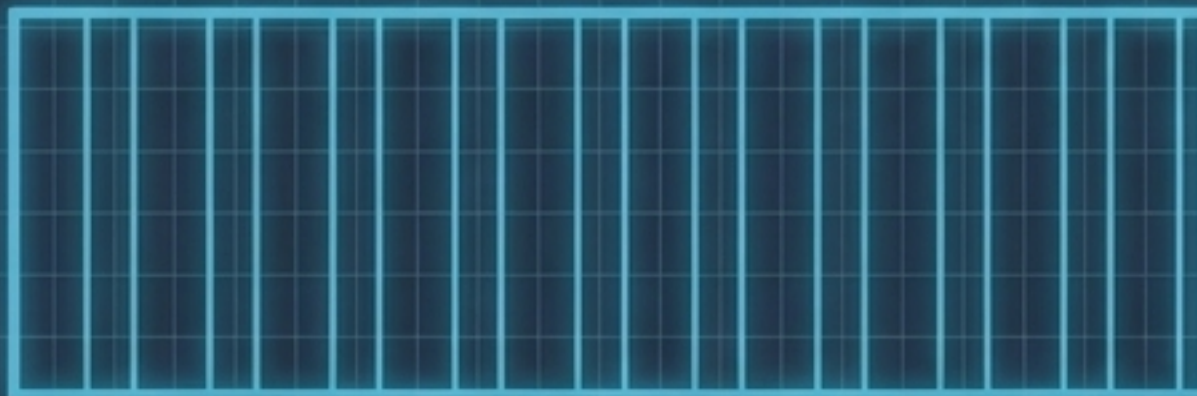


Coiling Geometry: Programming the Grain

Strands are laid onto a rotating mandrel. The helix angle (θ) determines whether the final product shreds like parallel muscle fibers or resists biting like a cross-hatched laminate.



Hoop Coiling ($\theta \approx 90^\circ$)



Maximum shred along grain.

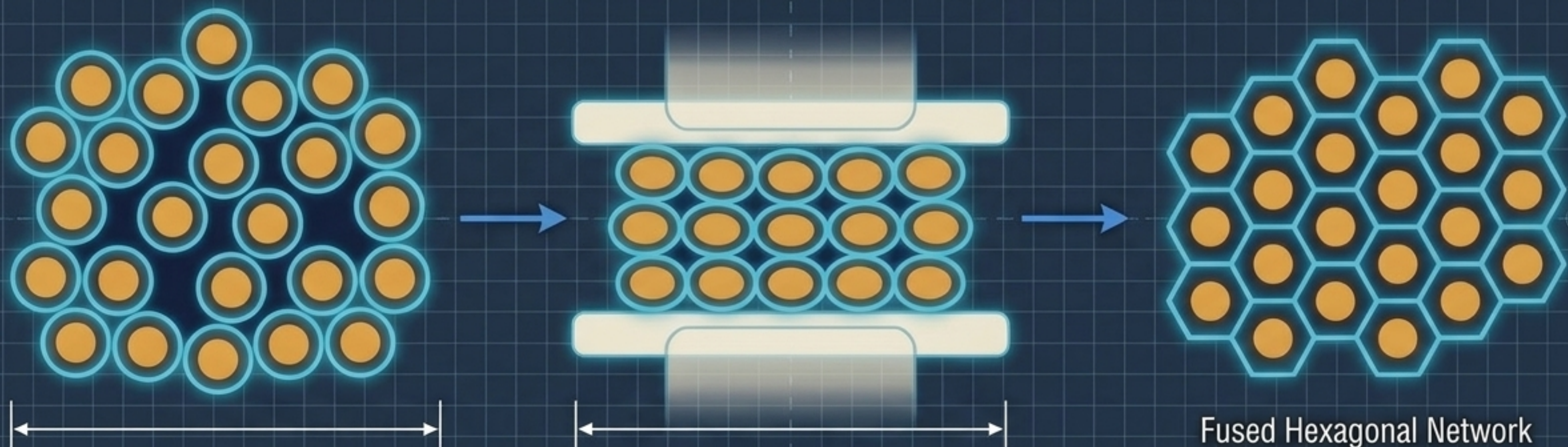
Crossed Layers ($\pm 45^\circ$)



Laminate resistance, multi-directional chew.

Consolidation: Forming the Network

- Hypothesis 4 (Shell Fusion): Under controlled pressure and temperature (P , T , t), adjacent shells deform and fuse into a continuous structural network, while cores remain isolated channels.



The Platform Control Panel

Separating inputs from outcomes allows precise targeting of mechanical properties.

Operator Setpoints (Inputs)

Formulation (C, S)



Preform Radii (r_0 , R_0)



Draw Ratio (λ)



Helix Angle (θ)



Press Conditions (P, T, t)



Resulting Architecture (Outputs)

8.0.0 | 0.0.0

Phase Chemistry

33.20% | ϕ

Core Fraction (ϕ)
[Juiciness & Fat]

160 microns | d

Strand Diameter (d)
[Fineness of Grain]



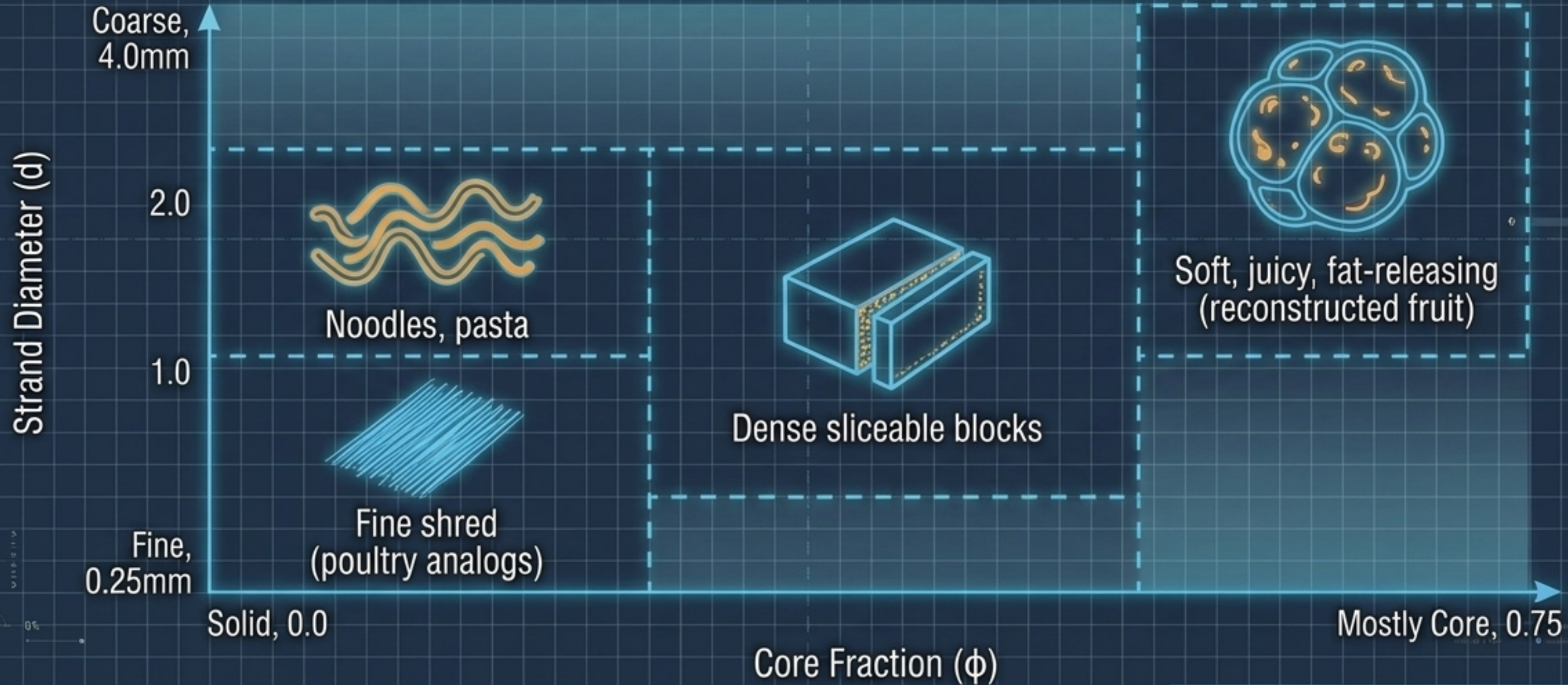
Orientation & Bite

0.00% | η
15.50 | f

Packing Fraction (η)
& Fusion Index (f)
[Shred vs. Solid]

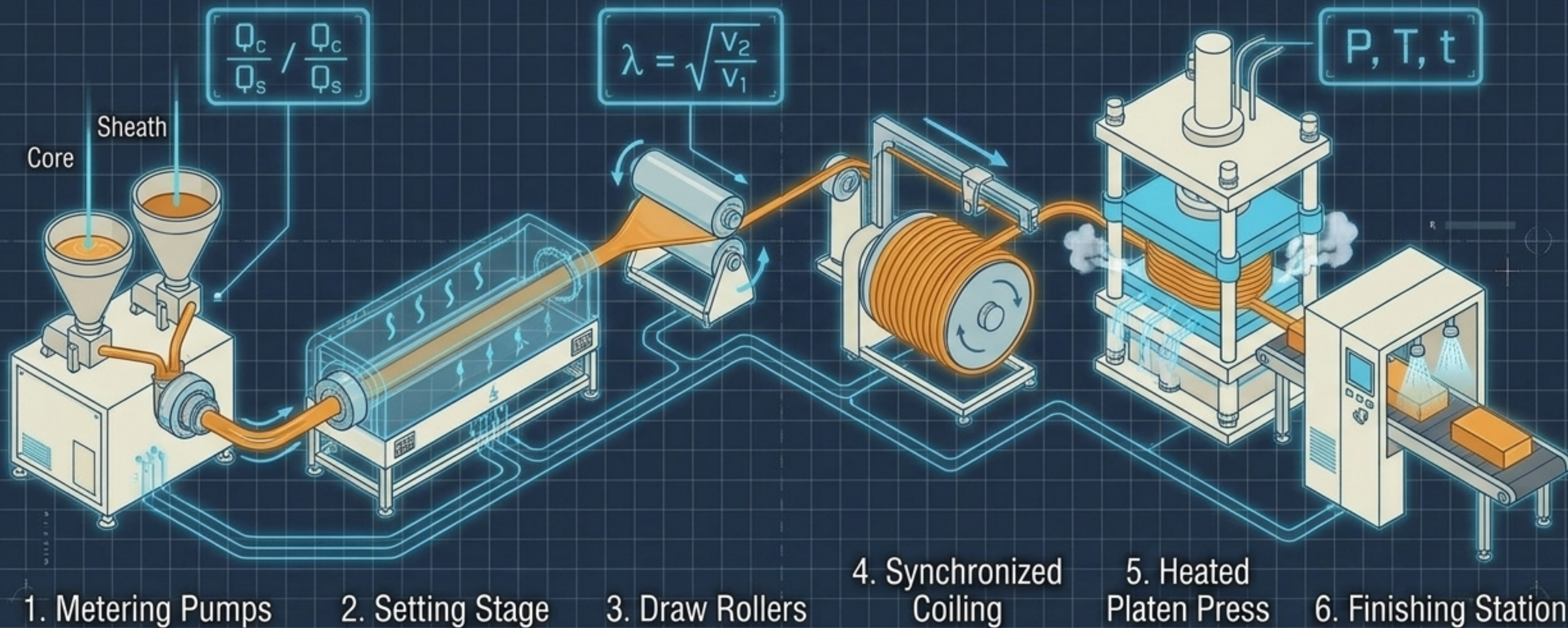
The Product Map: Traversing the Design Space

By holding helix angle (θ) and fusion (f) constant, sweeping through core fraction (ϕ) and strand diameter (d) unlocks distinct product categories on shared hardware.



Continuous Pilot Architecture

From fluid inputs to consolidated structures in a single synchronized sequence.



Availability Before Restriction

Environmental and ethical food transitions fail without genuinely acceptable substitutes.

Texture and flavor are fundamentally technical problems of processing, geometry, and chemistry.

The strand platform is not just a novel food form; it is a structural operating system designed to make sustainable substitutes mathematically scalable, customizable, and broadly available.



> INITIALIZE SEQUENCE